Finite Groups and Their Representations — Exercise Sheet 8

Induction and Frobenius groups

Variant: Questions only

Exercise 1 (Inducing from the identity subgroup). Let G be a finite group. Identify $\operatorname{Ind}_{\mathbf{1}}^{G}\mathbf{1}$ explicitly as a G-module. Use Frobenius reciprocity, rather than a character sum, to find the multiplicity of each $U \in \operatorname{Irr}(G)$ in it. Deduce the degree-square identity.

Exercise 2 (Induction in stages). Let $H \le K \le G$ be finite groups and let W be an H-representation. Prove an explicit isomorphism

$$\operatorname{Ind}_{K}^{G}\operatorname{Ind}_{H}^{K} W \cong \operatorname{Ind}_{H}^{G} W.$$

Write its inverse and check both balancing relations, not only dimensions.

Exercise 3 (The table of A_4 from a cyclic subgroup). Let $c = (1\,2\,3)$, $H = \langle c \rangle \le A_4$ and $\zeta = \exp(2\pi \mathrm{i}/3)$. Let $\theta_j \colon H \to \mathbb{C}^{\times}$ be the character with $\theta_j(c) = \zeta^j$ for $j = 0, 1, 2$. Let L_j be the lift from A_4/V_4 of the linear character taking cV_4 to ζ^j, with $L_0 = \mathbf{1}$. Let $U = \{x \in \mathbb{C}^4 : \sum_i x_i = 0\}$ with the natural A_4-action. Use induction from H to build the complete character table of A_4. Label its two classes of three-cycles by c and c^{-1}. Identify the three linear characters through $A_4/V_4 \cong C_3$.

Exercise 4 (Two different inductions from an involution). In A_4 put $a = (1\,2)(3\,4)$, $H = \langle a \rangle$, and $K = V_4$. Let ε be the nontrivial character of H. Decompose $\operatorname{Ind}_{H}^{A_4}\mathbf{1}$ and $\operatorname{Ind}_{H}^{A_4}\varepsilon$. For the second, also use induction in stages through K. Use the modules L_j and U of the preceding exercise, with $L_0 = \mathbf{1}$.

Exercise 5 (Dihedral planes by Mackey's criterion). Let $n \ge 3$, $H = \langle r \rangle \le D_n$, and $\theta_k(r) = \exp(2\pi \mathrm{i} k/n)$ for k modulo n. Use Mackey's criterion to decide when $\operatorname{Ind}_{H}^{D_n}\theta_k$ is irreducible. Find its generator matrices on $\{e, s\}$ and its splitting in the other cases. Give an example where θ_k is not faithful but its induction is irreducible. For the irreducible planes, determine when two parameters give isomorphic representations and show that these exhaust the irreducible planes of D_n.

Exercise 6 (A twist whose direction matters). Use the external semidirect product $C_7 \rtimes C_3$ from Sheet 3. Since $2^3 \equiv 1 \pmod 7$, it has the model $G = \{(a, b) : a \pmod 7,\ b \pmod 3\}$ with multiplication $(a, b)(c, d) = (a + 2^b c, b + d)$, computed modulo seven and three in the respective coordinates. Put $x = (1, 0)$, $y = (0, 1)$ and $H = \langle x \rangle$. For $\theta(x) = \zeta = \exp(2\pi \mathrm{i}/7)$, compute $\operatorname{Res}_{H}^{G}\operatorname{Ind}_{H}^{G}\theta$ with the ordered representatives e, y, y^2. Use Mackey's criterion to prove irreducibility and give both generator matrices.

Exercise 7 ⋆. Let $1 < H < G$ be finite with $H \cap gHg^{-1} = \mathbf{1}$ for every $g \notin H$. Such a pair gives a *Frobenius group*: in the transitive action on G/H, no nonidentity element fixes two points. Put $m = [G : H]$, $\psi = \chi_{\operatorname{Ind}_{H}^{G}\mathbf{1}}$, and let K consist of e and the elements fixing no coset. Prove Frobenius's theorem that K is a normal subgroup and $G = K \rtimes H$, by these steps.

(i) Show $N_G(H) = H$, count the conjugates of H, and prove $|K| = m$.

(ii) For each irreducible character θ of H of degree d, set $\widetilde{\theta} = \chi_{\mathrm{Ind}_H^G \theta} - d(\psi - 1_G)$. Show that it equals $\theta(h)$ on $h \in H \setminus \{e\}$ and equals d on all of K. Extend the first description by conjugacy.

(iii) Show $\widetilde{\theta}$ is an integral combination of irreducible characters, has squared norm one and positive identity value, hence is an irreducible character by Lecture 6.

(iv) Use separation of each nonidentity element from the identity by irreducible characters of H to show $K = \bigcap_\theta \ker \widetilde{\theta}$. Here θ ranges over all irreducible characters of H. Conclude normality and the semidirect product.